

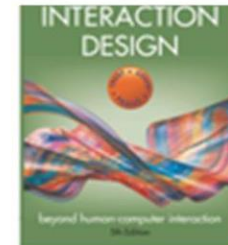


**IN
PARTNERSHIP
WITH
PLYMOUTH
UNIVERSITY**

PUSL3122 HCI, Computer graphics and visualisation

Topics covered last lecture (Lecture 3)

- **Evaluation**: key concepts, terms, different types of evaluation methods. Different evaluation methods, Hybrid approach is best.
- **Practice**: how to do usability testing, outline the basics of experimental design, how to do field studies
- **Inspection**: evaluation without users



Chapter 14
INTRODUCING
EVALUATION



Chapter 15
Evaluation Studies

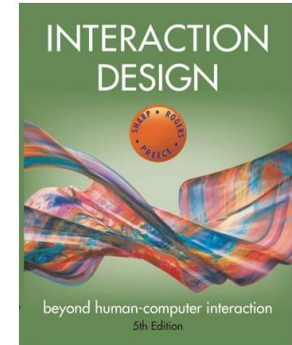


Chapter 16
Evaluation

Topics covered this lecture (Lecture 4)...

- **Interfaces**

- Metaphors, interface types, choosing interfaces to match user needs
- This part of the module is taken from the main text book for the module, the latest release from 2019. This lecture focuses on chapter 7.



Chapter 7 INTERFACES

20 interface types covered



1. Command
2. Graphical
3. Multimedia
4. Virtual reality
5. Web
6. Mobile
7. Appliance
8. Voice
9. Pen
10. Touch
11. Gesture
12. Haptic
13. Multimodal
14. Shareable
15. Tangible
16. Augmented Reality
17. Wearables
18. Robots and drones
19. Brain-computer interaction
20. Smart

Second Life command line-based interface for visually-impaired users



Graphical user interfaces (GUIs)

- **Xerox Star** first **WIMP** gave rise to GUIs 😊
 - Windows
 - Sections of the screen that can be scrolled, stretched, overlapped, opened, closed, and moved around the screen using the mouse
 - Icons
 - Pictograms that represent applications, objects, commands, and tools that were opened when clicked on
 - Menus
 - Lists of options that can be scrolled through and selected
 - Pointing device
 - A mouse controlling the cursor as a point of entry to the windows, menus, and icons on the screen

Example of first generation GUI

The screenshot shows a window titled "Recurring Tasks". It contains several input fields and controls:

- Description:** A text input field.
- Default Priority:** A small square input field.
- Notes:** A large text area with scrollbars.
- Est. Time Blocks:** A text input field containing "0".
- Category:** A text input field containing "None".
- Start:** A date/time input field showing "01 .01 .80" with up/down arrows and a calendar icon.
- End:** A date/time input field showing "01 .01 .99" with up/down arrows and a calendar icon.
- Day:** A text input field containing "0".
- Month:** A text input field containing "0".
- Weekdays:** A group box containing checkboxes for Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, and Sunday.
- Weeks:** A group box containing checkboxes for First, Second, Third, Fourth, Fifth, and Last.
- Every:** A text input field containing "0".
- Days:** A radio button.
- Months:** A radio button.
- Weeks:** A radio button.
- Buttons:** "Save", "Exit", "New", "Delete", "Find", and "Help".
- Calendar:** A calendar view showing the days of the week (Sun, Mon, Tue, Wed, Thu, Fri, Sat) and a large empty area for the month.
- Performance:** A checkbox labeled "Performance".
- Req:0**, **Comp:0**, **%:0**: Three text labels at the bottom right.

Window design

- Windows were invented to overcome the **physical constraints** of a computer **display**
 - They enable **more information** to be viewed and tasks to be performed
- **Scroll bars** within windows enable **more information** to be viewed
- **Multiple windows** can make it difficult to find desired one
 - **Listing, tabbing, and thumbnails** are techniques that can help

Menu styles

- **Flat list**: Good for showing large number of options at the same time when display is small
- **Drop down**: Shows more options on same screen (for example, cascading)
- **Pop-up**: When pressed, command key for relevant options
- **Contextual**: Provides access to often-used commands associated with a particular item
- **Collapsible**: **Toggles between + and – icons on a header to expand or contract its contents**
- **Mega**: All options shown using 2D drop-down layout

Template for a collapsible menu

Section 1

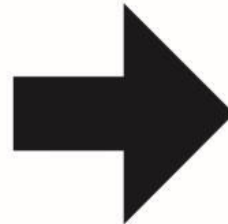
Section 1 content line 1
Section 1 content line 2
Section 1 content line 3
Section 1 content line 4

Section 1 content line 5
Section 1 content line 6
Section 1 content line 7
Section 1 content line 8

Section 2

Section 2 content line 1
Section 2 content line 2
Section 2 content line 3
Section 2 content line 4

Section 2 content line 5
Section 2 content line 6
Section 2 content line 7



Section 1

Section 2

Section 3

Section 4

Section 5

A mega menu

JOHN LEWIS & PARTNERS

Search product or brand

Q

Create an account

Sign in







Home & Garden

Furniture & Lights

Electricals

Women

Men

Beauty

Baby & Child

Sport & Leisure

Gifts

Offers

Furniture by Room

Bedroom

Living Room

Dining Room

Kitchen

Bathroom

Children's

Nursery

Home Office

Hallway

Garden

Furniture by Type

Beds

Mattresses

Wardrobes

Bedside Tables

Chests of Drawers

Dressing Tables

Headboards

Bookcases, Shelving Units & Shelves

Office Desks

Office Chairs

Bathroom Cabinets

Sofas & Armchairs

Sofa Beds

Coffee Tables

Side Tables

Console Tables

TV Stands

Cab nets & Sideboards

Dining Tables

Dining Chairs

Dining Furniture Ranges

Bar Chairs & Stools

Lighting

View all Lighting

Ceiling Lighting

Wall Lighting

Desk & Table Lamps

Floor Lamps

Ceiling & Lamp Shades

Line, Fairy & Novelty Lights

SAD & Wake Up Lighting

Cabinet Lighting

Light Bulbs

Lighting & Electrical Accessories

Lighting By room

Garden & Outdoor Lighting

Bathroom Lighting

Kitchen Lighting

Children's Lighting

Don't Miss

New in: Lighting

New in: House by John Lewis

New in: Made to Order Upholstered Beds

Bring the outdoors in: Conservatory Furniture

Try our new Sofa Discovery Tool

20% off Mattresses

20% off Sofas & Armchairs

20% off Living & Dining Furniture

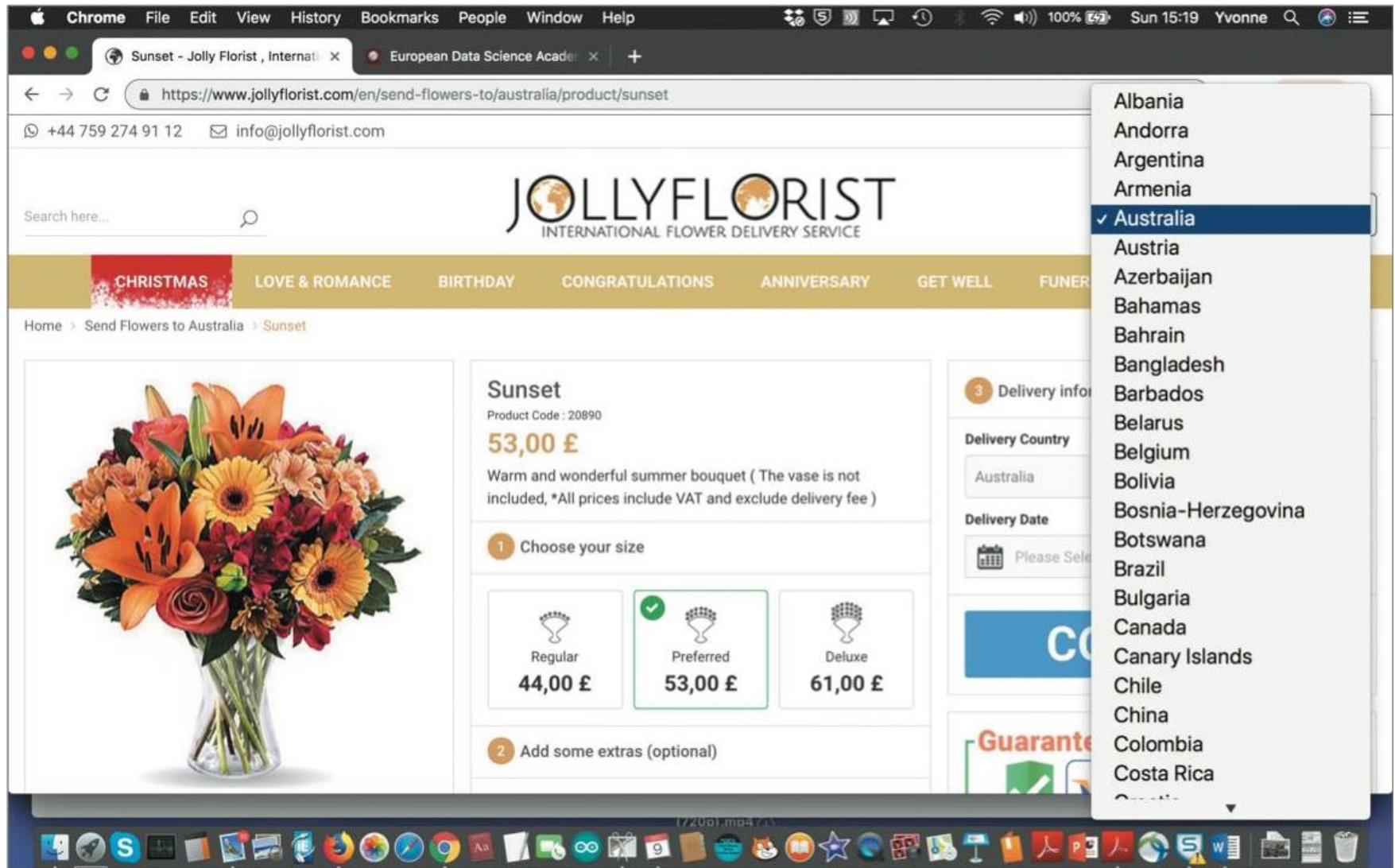
20% off Fitted Bedrooms

30% off Special Buy Fitted Kitchens

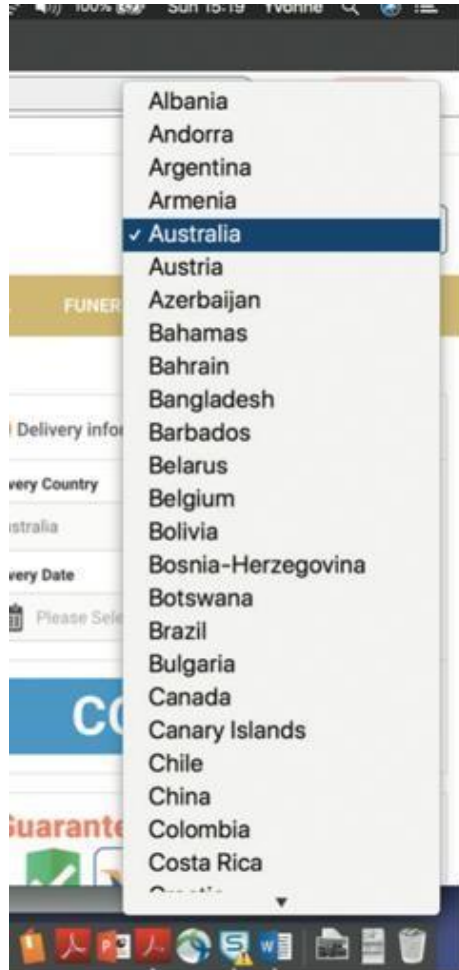
Research and design considerations

- Window management
 - Enables users to **move fluidly** between different **windows** (and monitors)
- How to **switch attention** between windows without getting distracted
- Design principles of **spacing, grouping, and simplicity** should be used
- Which **terms** to use for menu options (for example, “front” versus “**bring to front**”)
- **Mega** menus easier to navigate than drop-down ones

Selecting a country from a scrolling window



Selecting a country ...



What could we do
to **improve this**?

Is this method any better?

F	G	H	I	J
Fiji Finland France French Guyana French Polynesia	Gabon Germany Gibraltar Greece Greenland Guadeloupe Guam Guatemala	Haiti Holland Honduras Hong Kong Hungary	Iceland India Indonesia Iran Ireland Israel Italy Ivory Coast	Jamaica Japan Jordan

Icon design

- Icons are **assumed** to be easier to learn and remember than commands
- Icons can be designed to be **compact** and **variably** positioned on a screen
- Now pervasive in every interface
 - For example, they represent desktop objects, tools (for example, a paintbrush), applications (for instance, a web browser), and operations (such as cut, paste, next, accept, and change)

Simple smartwatch menus with 1, 2, or 3 options



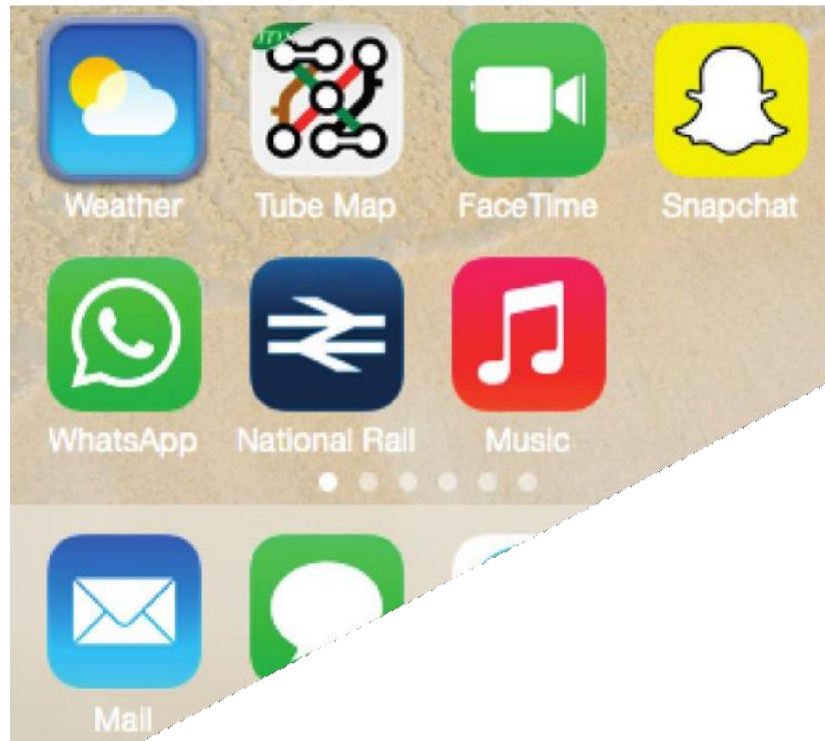
Icons

- Since the **Xerox Star** days, icons have changed in their look and feel:
 - black and white
 - Now we have colour, shadowing, photorealistic images, 3D rendering, and animation
- Many designed to be **very detailed** and **animated** making them both visually attractive and informative
- Can be highly inviting, **emotionally appealing**, and feel alive

Icon forms

- The **mapping** between the **representation** and underlying **referent** can be:
 - **Similar**
 - For example, a picture of a file to represent the object file
 - **Analogical**
 - For instance, a picture of a pair of scissors to represent 'cut'
 - **Arbitrary**
 - such as the use of an X to represent 'delete'
- The most effective icons are similar ones
- Many operations are actions making it more difficult to represent them
 - Use a combination of objects and symbols that capture the salient part of an action

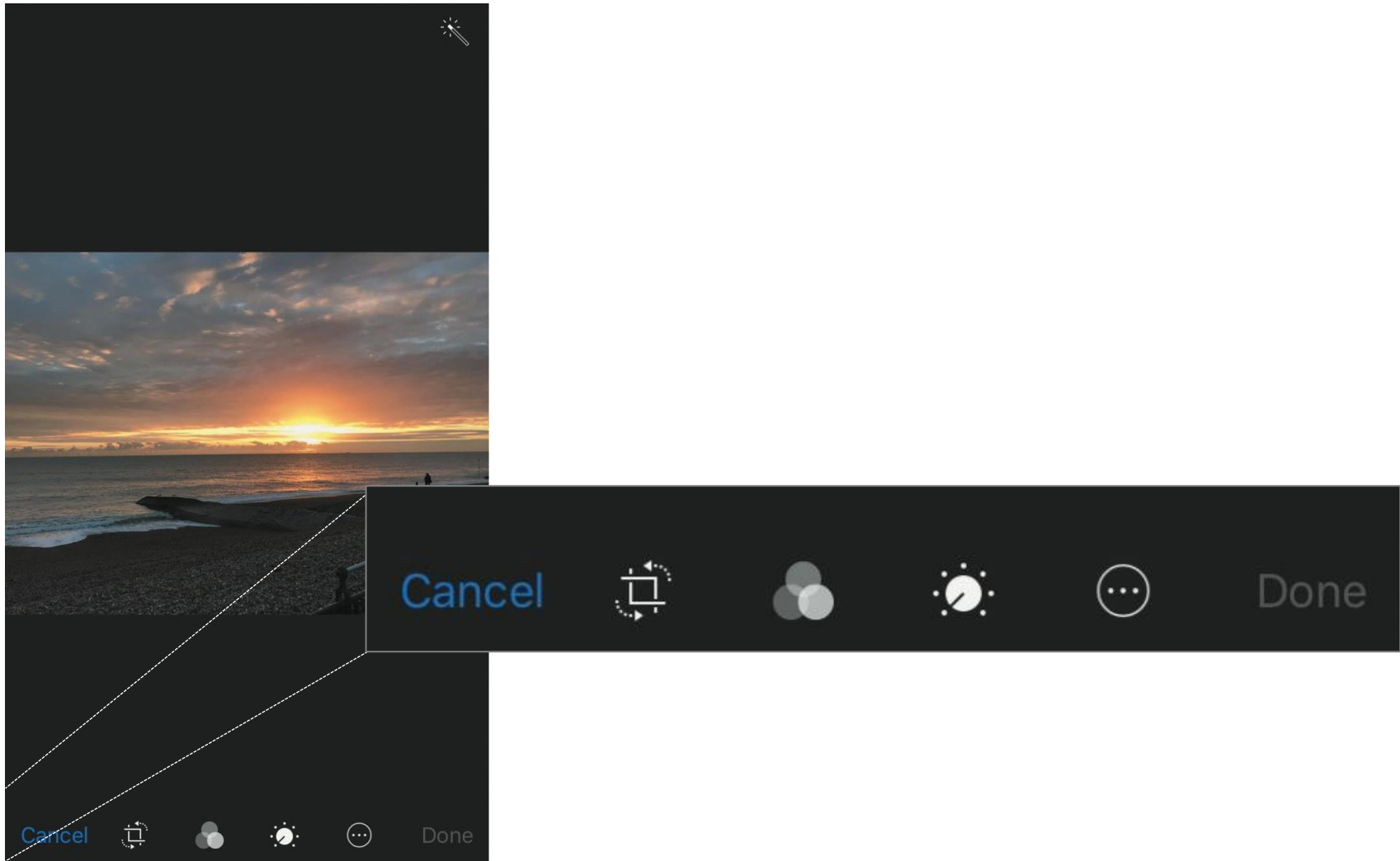
Flat 2D icons for a smartphone and a smartwatch



What does this icon mean?



Basic edit icons that appear on the iPhone app





What changed?

Research and design considerations

- There is a **wealth of resources** for creating icons
 - Guidelines, style guides, icon builders, libraries, online tutorials
- **Text labels** can be used **alongside icons** to help identification for small icon sets
- For large icon sets (for instance, photo editing or word processing) can **use the hover function** to stop cluttering up the interface

Multimedia

- **Combines different media** within a single interface with various forms of interactivity
 - Graphics, text, video, sound, and animation
- Users click on **links in an image or text**
 - Another part of the program
 - An animation or a video clip is played
 - Users can return to where they were or move on to another place
- **Can** provide better ways of presenting information than a single media can – but has to be done carefully and consistently

Multimedia learning app designed for tablet



Clearly, this is a **richer** learning experience

Virtual reality

- Computer-generated **graphical simulations** providing:
 - “the **illusion** of participation in a **synthetic** environment rather than external observation of such an environment”

Gigante, M. A. (1993) Virtual Reality: Enabling Technologies. In R. A. Earnshaw, M. A. Gigante and H. Jones (eds) *Virtual Reality Systems*. Academic Press, London, pp. 15–25.

- Provide **new** kinds of experience, enabling users to interact with objects and navigate in 3D space
- Create **highly-engaging user experiences**
- Long way to go with these as well

Application areas

- Video games
- Arcade **games** for social groups
- **Therapy** for fears
- Experience how others feel emotions
 - For example, **empathy** and **compassion**
- Enrich user's planning experience for **travel** destinations
- Architecture, design, and education
- **BEST** = Patients who are dying “run with wild horses in Iceland, traverse the Java sea, go skydiving and tour Venice's canals aboard a gondola, all without leaving their beds.”

www.death.io/can-virtual-reality-help-those-who-are-dying/

Website design

- Early websites **were largely text-based**, providing hyperlinks
- Concern was with how best to **structure information** to enable users to **navigate** and access them easily and quickly
- Nowadays, more emphasis is on making pages distinctive, **striking**, and **aesthetically** pleasing
- Need to think of how to design information for **multiple platforms**—keyboard or touch?
 - For example, smartphones, tablets, and PCs

Usability versus aesthetics?

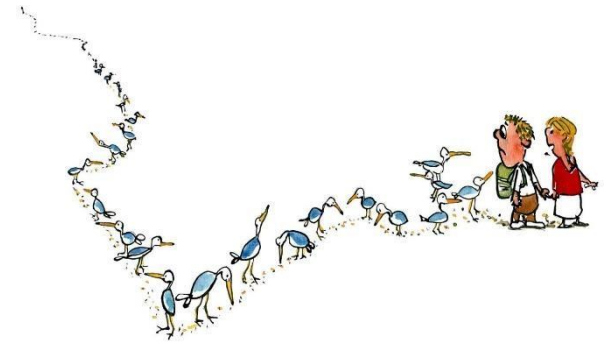
- **Vanilla or multi-flavor** design?
 - Ease of finding something versus aesthetic and enjoyable experience
- Web designers are “**thinking great literature**”
- **Users read the web like a “billboard going by at 60 miles an hour”**

Krug, S. (2014) *Don't Make Me Think, Revisited: A Common Sense Approach to Web Usability* (3rd edn). Pearson.

- Need to determine **how** to brand a web page to catch and keep ‘eyeballs’

Breadcrumbs for navigation

- **Breadcrumbs** are category labels:
 - Enable users to look at other pages without losing track of where they have come from
 - Very usable
 - Enable one-click access to higher site levels
 - Attract first time visitors to continue to browse a website having viewed the landing page



Hansel, Gretel and the Breadcrumb Birds

<https://fritsahlefeldt.com/2012/04/25/hansel-gretel-and-the-breadcrumb-birds/>



Research and design considerations

- Many books and guidelines on website design
- Veen's three core questions to consider when designing any website:
 - **Where** am I?
 - Where **can I go**?
 - **What's here**?

Veen, J. (2001) *The Art and Science of Web Design*. New Riders Press, Indianapolis, IN.

Mobile interfaces

- **Handheld** devices intended to be used while on the move
- Have become **pervasive**, increasingly used in all aspects of everyday and working life
 - For example, phones, fitness trackers, and smartwatches
- **Larger-sized tablets** used in mobile settings
 - Including those used by flight attendants, marketing professionals, and at car rental returns

QR codes and smartphones



Great way to sign up for more information ...

Research and design considerations

- Mobile interfaces can be **cumbersome** to use for those with **poor manual dexterity (ME!) or 'fat' fingers**
- Key concern is **hit area**:
 - Area on the phone display that the user touches to make something happen, such as a key, an icon, a button, or an app
 - Space needs to be **big enough** for all fingers to press accurately
 - If too small, the **user may accidentally** press the wrong key
 - Fitts' law can be used to help design right spacing.
 - Minimum tappable areas should be 44 points x 44 points for all controls

<https://www.interaction-design.org/literature/article/fitts-s-law-the-importance-of-size-and-distance-in-ui-design>

Appliances

- Everyday devices in home, public places, or car
 - For example, washing machines, remotes, toasters, printers, and navigation systems)
- And personal devices
 - For instance, digital clock and digital camera
- Used for short periods
 - For example, starting the washing machine, watching a program, buying a ticket, changing the time, or taking a snapshot
- **Need to be usable** with **minimal**, if any, **learning**

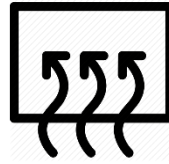


Simple toaster control



But sometimes they aren't

What does this icon represent?



Is the Front
or back
switched on?



Where do you think the
confusion came from?



But sometimes they aren't

Some solutions added text ?



Are front
windcreens
that shape?



Now they are placed next
to each other



Any comments?

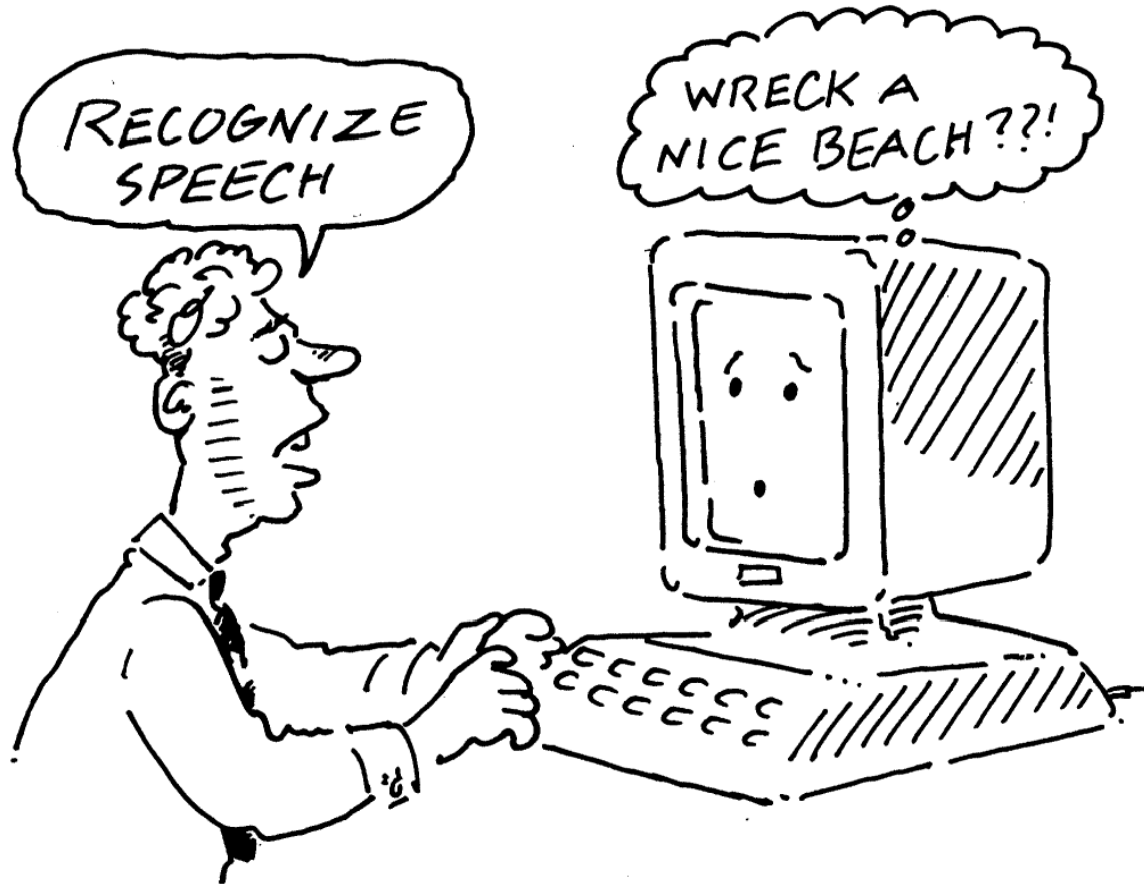
Research and design considerations

- Need to design as **transient** (lasting only for a short time) interfaces with short interactions
- **Simple** interfaces
- Consider **trade-off between soft and hard** controls
 - For example, use of buttons or keys, dials, or scrolling

Voice User Interfaces

- Involves a **person talking** with a spoken language app, for example, timetable, travel planner, or phone service
- Used most for inquiring about specific information, for example, flight times or to perform a transaction, such as buying a ticket
- Also used by **people with visual impairments**
 - For example, speech recognition word processors, page scanners, web readers, and home control systems

Have speech interfaces come of age?



Modeling human conversations

- People **often interrupt** each other in a conversation
 - (interesting as a result of all our zoom use)
 - Especially when ordering in a restaurant, rather than let the waiter go through all the options
- Speech technology has a similar feature called ‘barge-in’
 - Users can choose an option before the system has finished listing all the options available
- It’s a really difficult task ...

Voice assistants (for example, Alexa)

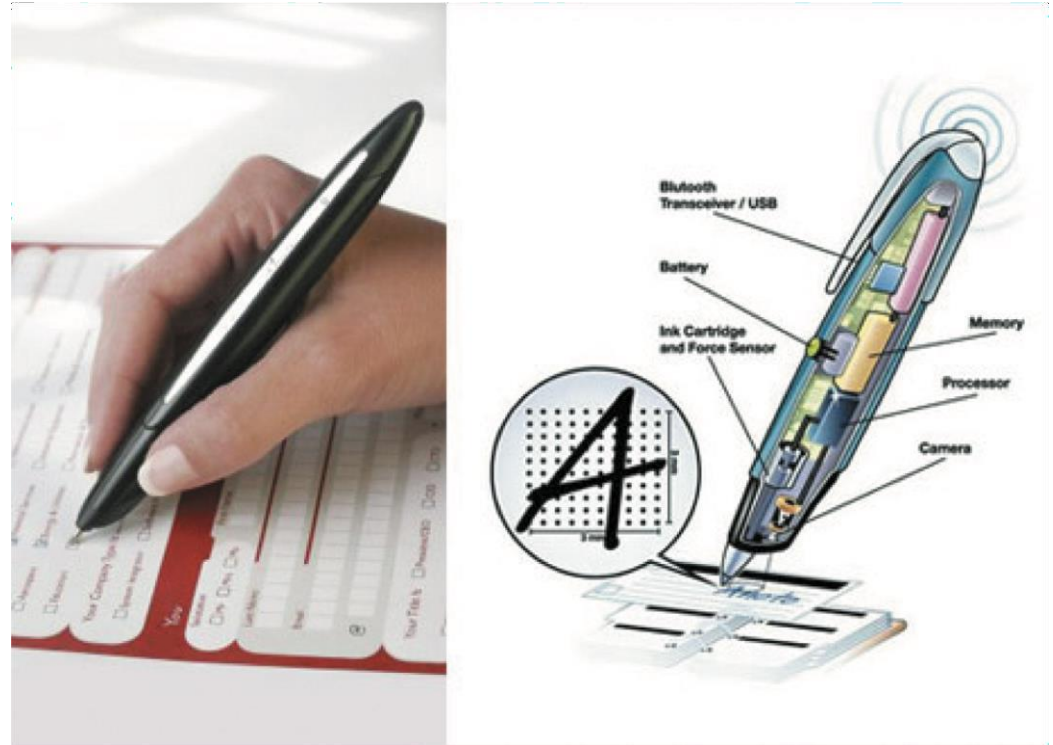
- Have **become popular** in many homes
- Allow all to use rather than being single use
- Support **families** playing games, interactive storytelling, jokes, and so forth
- Can encourage social and emotional bonding
- Young children (under 4), however, find it **difficult to be understood** by the voice assistants
 - Frustrating for them

Research and design considerations

- How to design systems that can keep conversation on track
 - **Help people navigate** efficiently through a menu system
 - **Enable them to recover** easily from errors
 - Guide those who are vague or ambiguous in their requests for information or services
- Type of voice actor (for example, male, female, neutral, or dialect)
 - **Do people prefer to listen to and are more patient with a female or male voice, a northern or southern accent?**

Pen-based devices

- **Enable people to write**, draw, select, and move objects at an interface using light pens or styluses
 - Capitalize on the well-honed drawing skills developed from childhood
- Digital ink, for example, **Anoto**, use a combination of ordinary ink pen with digital camera that digitally records everything written with the pen on special paper

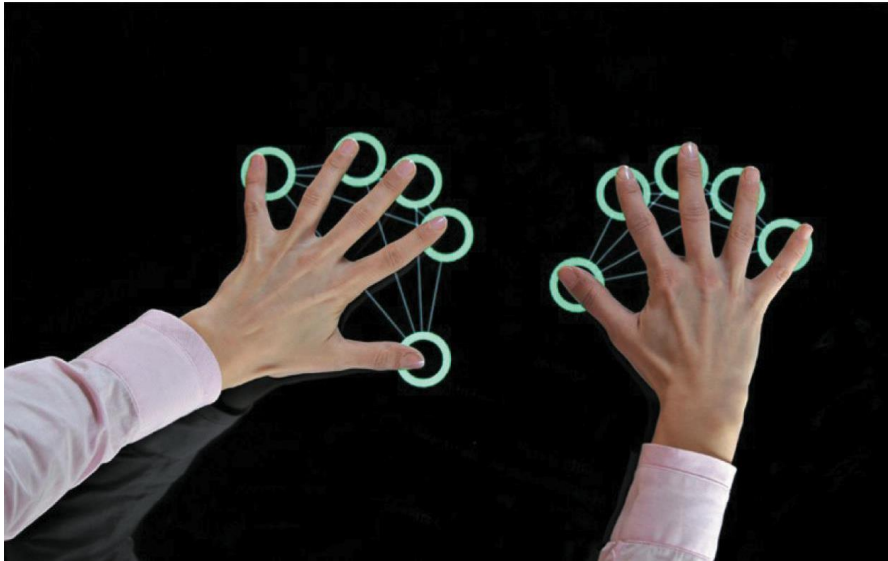


<https://www.anoto.com/cases/anoto-digital-pen/>

Touchscreens

- **Single touchscreens** are used in walk-up kiosks (such as ticket machines and ATMs) to detect the presence and location of a person's touch on the display
- **Multi-touch surfaces** support a range of more dynamic finger tip actions, for example, swiping, flicking, pinching, pushing, and tapping
- They do so by registering **touches at multiple locations** using a grid
- Now used for many kinds of displays, such as smartphones, iPods, tablets, and tabletops
 - Supports one and two hand gestures, including tapping, zooming, stretching, flicking, dwelling, and dragging

A multi-touch surface (that's 10 inputs!)



https://www.youtube.com/watch?v=89sz8ExZndc&ab_channel=trevz0r

Jeff Han back in 2006
Slow progress! Why?

Research and design considerations

- Provides **fluid** and **direct styles of interaction** involving freehand and pen-based gestures for certain tasks
- Core design concerns include whether **size, orientation,** and **shape of touch** displays effect collaboration
 - Much **faster to just scroll through wheels**, carousels, and bars of thumbnail images or lists of options by finger flicking
 - **New gestures need to be learned** for multi-touch, so a small set of gestures for common commands is preferable
 - **More cumbersome**, error-prone, and slower to type using a virtual keyboard on a touch display than using a physical keyboard
- They will **evolve** into more usable products ...

Gesture-based systems

- **Gestures** involve moving arms and hands to communicate
- Uses camera recognition, sensor, and computer vision techniques
 - Recognize people's arm and hand **gestures** in a room
 - Gestures need to be presented sequentially to be understood (compare with the way sentences are constructed)



Recognizes core gestures for manipulating MRI or CT images using Microsoft Kinect

Haptic interfaces

- Provide **tactile** feedback
 - By applying **vibration** and **forces** to a person's body, using actuators that are embedded in their clothing or a device they are carrying, such as a smartphone
- **Vibrotactile** feedback can be used to **simulate the sense of touch** between remote people who want to communicate
 - good use = patients alone in covid wards
- **Ultrahaptics** creates the illusion of touch in midair using ultrasound to make the illusion of 3D shapes



Provides **nudges** when playing violin incorrectly

Research and design considerations

- Where best to **place actuators** on body
- Whether to use **single or sequence of 'touches'**
- When to **buzz** and **how** intense
- How does the wearer feel it in **different contexts**?
- What kind of **new smartphone/smartwatch apps** can use vibrotactile creatively?
 - For example, **slow tapping to feel like water drops meant to indicate that it is about to rain**, and heavy tapping to indicate a thunderstorm is looming

Multimodal Interfaces- Tracking a person's movements



- **Kinect camera** can detect **multimodal input** in real time using RGB camera for facial recognition and gestures, depth camera for movement tracking, and microphones for voice recognition
- Used to **build model of person** and **represented as avatar** on display programmed to move just like them

Shareable interfaces

- Designed for **more than one person** to use:
 - Provide **multiple inputs** and sometimes allow simultaneous input by co-located groups
 - **Large wall displays** where people use their own pens or gestures
 - **Interactive tabletops** where small groups interact with information using their fingertips
 - For example, DiamondTouch, Smart Table, and Surface



(a)



(b)

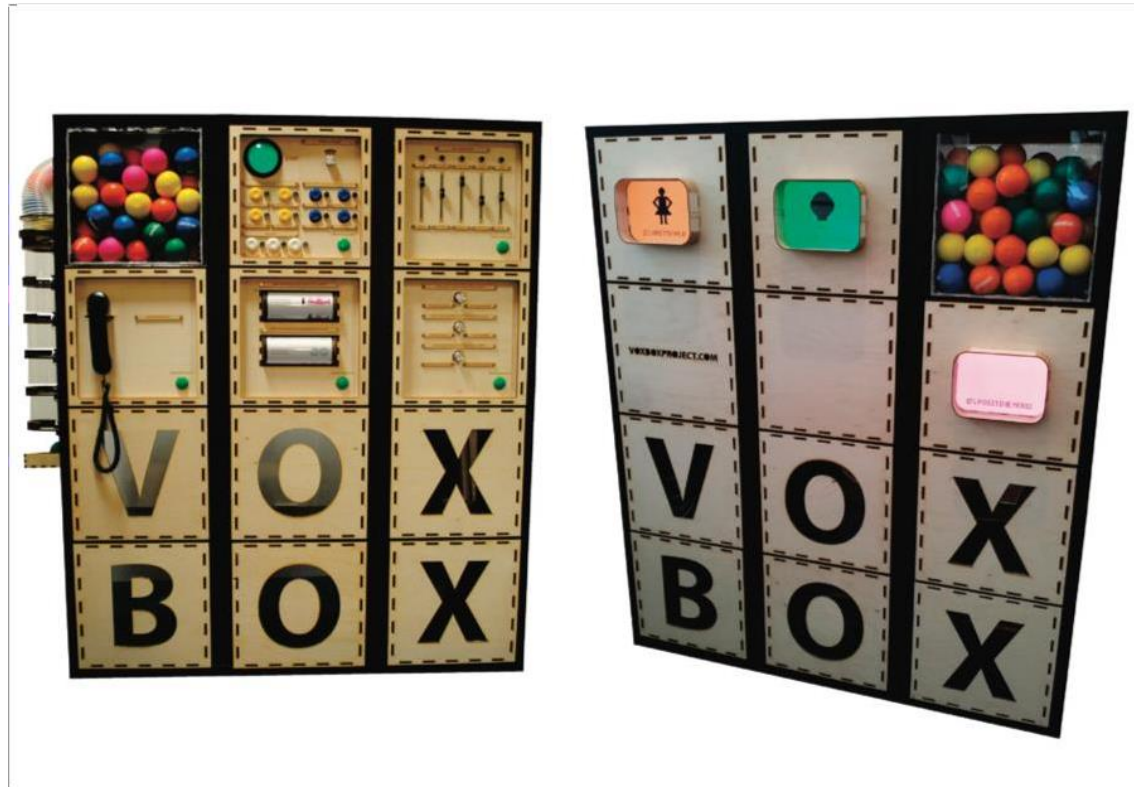
Tangible Interfaces

- Type of **sensor-based interaction**, where physical objects, for example, bricks, are coupled with digital representations
- When a **person manipulates the physical object/s**, it causes a digital effect to occur, for example, an animation
- **Digital effects** can take place in a number of media and places, or they can be **embedded** in the physical object

VoxBox

- A tangible system that gathers opinions at events through playful and engaging interaction

Golsteijn, C., Gallacher, S., Koeman, L., Wall, L., Andberg, S., Rogers, Y., and Capra, L. (2015) VoxBox: a Tangible Machine That Gathers Opinions from the Public at Events. In *Proc. of TEI 2015*. ACM.



Augmented Reality

- Augmented reality: **Virtual representations** are **superimposed** on physical devices and objects
- **Pokémon Go** made it a household game
 - Used smartphone camera and GPS to place virtual characters onto objects in the environment as if they really are there
- Many other applications including medicine, navigation, air traffic control, games, and everyday exploring
- In medicine
 - Virtual objects, for example, x-rays and scans, are overlaid on part of a patient's body
 - Aid the physician's understanding of what is being examined or operated

Augmented reality overlay on a car windshield



Wearables

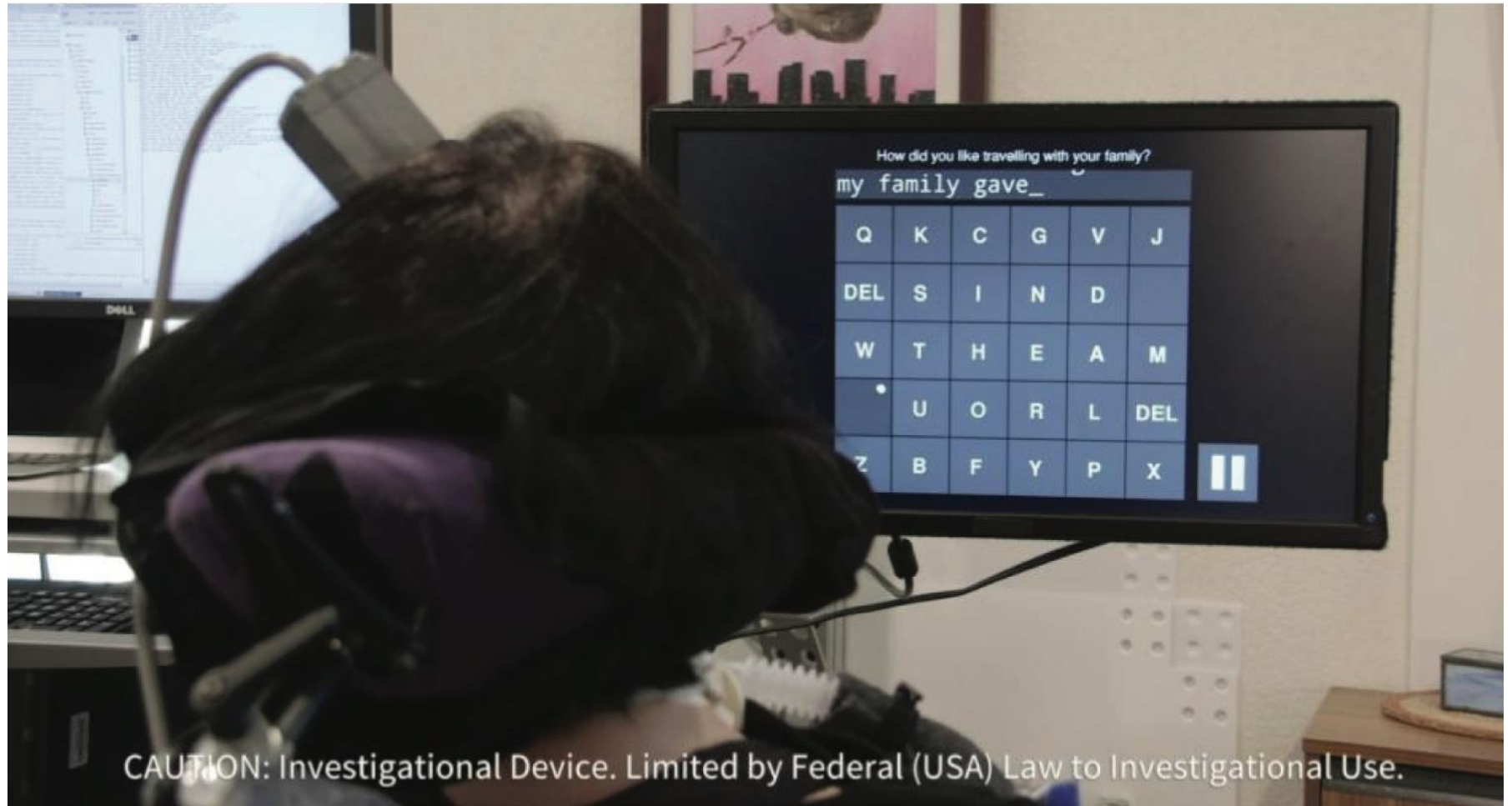
- First developments were **head- and eyewear-mounted cameras** that enabled user to record what was seen and to access digital information
- Since then, jewelry, head-mounted caps, smart fabrics, glasses, shoes, and jackets have all been used
 - Provides the user with a means of interacting with digital information while on the move
- Applications include automatic diaries, tour guides, cycle indicators, and fashion clothing



Google Glass (2014). Why was there so much excitement and concern about people filming what they could see right in front of them?

- Comfort (light, small)
- Hygiene (wash)
- Easy to wear
- Usability

A **brain-computer interface** being used by a woman who is paralyzed to select letters on the screen



Robots

- Main types
 - Remote robots used in **hazardous** settings
 - Can be controlled to investigate bombs and other dangerous materials
 - **Domestic** robots helping around the house
 - Can pick up objects and do daily chores like vacuuming
- **Pet** robots as human companions
 - Have therapeutic qualities, helping to reduce stress and loneliness
- **Sociable** robots that work collaboratively with humans
 - Encourage social behaviors



<https://www.bbc.co.uk/news/uk-northern-ireland-25313668>



<https://phys.org/news/2014-08-paro-robot-elderly.html>

Drones

- **Unmanned aircraft** that are **controlled remotely** and used in a number of contexts
 - For example, **entertainment**, such as carrying drinks and food to people at festivals and parties
- **Agricultural** applications, such as flying them over vineyards and fields to collect data about crops, which is useful to farmers
 - Helping to **track poachers** in wildlife parks in Africa
 - Can **fly low and stream photos** to a ground station where images can be stitched together into maps - determine the health of a crop, or when it is the best time to harvest the crop



Drone being used to survey the state of a vineyard

Smart interfaces

- Smart: phones, speakers, watches, cars, buildings, cities
 - Smart refers to **having some intelligence** and connected to the internet and other devices
- **Context-aware**
 - Understand what is happening around them and execute appropriate actions, for example, a Nest thermostat
- **Human-building interaction**
 - Buildings are designed to sense and act on behalf of the inhabitants but also allow them to have some control and interaction with the automated systems

Thank You